

Lecture-#0

Natural Language Processing

A8706

Introduction

Syllabus Details



Learning from Experience, Sharing with Passion

Introduction: What is Natural Language Processing (NLP), Origins of NLP, The Challenges of NLP, Phases of NLP, Language and Grammar. Finding the Structure of Words and Documents: Words and Their Components, Issues and Challenges, Morphological Models. Finding the Structure of Documents: Introduction, Sentence Boundary Detection, Topic Boundary Detection, Methods, Complexity of the Approaches, Performances of the Approaches, Features, Processing Stages.

Syntax: Parsing Natural Language, A Data-Driven Approach to Syntax, Stop words, Correcting Words, Stemming, Lemmatization, Parts of Speech (POS) Tagging, Representation of Syntactic Structure, Parsing Algorithms, Models for Ambiguity Resolution in Parsing.

Syllabus Details



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Semantic Parsing: Introduction, Semantic Interpretation: Structural Ambiguity, Entity and Event Resolution, System Paradigms, Word Sense, Predicate-Argument Structure, Meaning Representation.

Language modelling: Introduction, n-Gram Models, Language Model Evaluation, Parameter Estimation, Types of Language Models: Class-Based Language Models, MaxEnt Language Models, Neural Network Language Models Language-Specific Modeling Problems, Multilingual and Crosslingual Language Modeling.

Applications: Question Answering: History, Architectures, Question Analysis, Search and Candidate Extraction, Automatic Summarization: Approaches to Summarization, Spoken Dialog Systems: Speech Recognition and Understanding, Speech Generation, Dialog Manager, Voice User Interface, Information Retrieval: Document Preprocessing, Monolingual Information Retrieval

Syllabus Details Lab

1.
 - a. Write a program to Tokenize Text to word using NLTK.
 - b. Write a program to Tokenize Text to Sentence using NLTK.
2.
 - a. Write a program to remove numbers, punctuations, and whitespaces in a file.
 - b. Write a program to Count Word Frequency in a file.
3. Write a program to Tokenize and tag the following sentence using Morphological Analysis in NLP.
4.
 - a. Write a program to get Synonyms from WordNet.
 - b. Write a program to get Antonyms from WordNet.
5.
 - a. Write a program to show the difference in the results of Stemming and Lemmatization.
 - b. Write a program to Lemmatizing Words Using WordNet.
6.
 - a. Write a program to print all stop words in NLP.
 - b. Write a program to remove all stop words from a given text.

Syllabus Details Lab



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7. Write a Python program to apply Collocation extraction word combinations in the text. Collocation examples are “break the rules,” “free time,” “draw a conclusion,” “keeps in mind,” “get ready,” and so on.
8. Write a Python program to extract Relationship that allows obtaining structured information from unstructured sources such as raw text. Strictly stated, it is identifying relations (e.g., acquisition, spouse, employment) among named entities (e.g., people, organizations, locations). For example, from the sentence “Mark and Emily married yesterday,” we can extract the information that Mark is Emily’s husband.
9. Write a program to print POS and parse tree of a given Text.
10. Write a program to print bigram and Trigram of a given Text.
11. Implement a case study of NLP application.

Reference Books

Text Books:

- Daniel M. Bikel Imed Zitouni., Multilingual Natural Language Processing Applications: From Theory to Practice, IBM Press, 2013.
- Tanveer Siddiqui, U.S. Tiwary., Natural Language Processing and Information Re- trieval, Oxford University, 2008.

Reference Books:

- Daniel Jurafsky and James H Martin., Speech and Language Processing: An introduction to Natural Language Processing, Computational Linguistics and Speech Recognition,2nd Edition, Prentice Hall, 2008.
- James Allen., Natural Language Understanding,2nd Edition, Cummings publishing company,1995.

Courses Outcomes

- ❖ A8706.1 Identify the structure of words and documents for text preprocessing.
- ❖ A8706.2 Choose an approach to parse the given text document.
- ❖ A8706.3 Make use of semantic parsing to capture real meaning of text.
- ❖ A8706.4 Select a language model to predict the probability of a sequence of words.
- ❖ A8706.5 Examine the various applications of NLP.

Introduction to Natural Language Processing

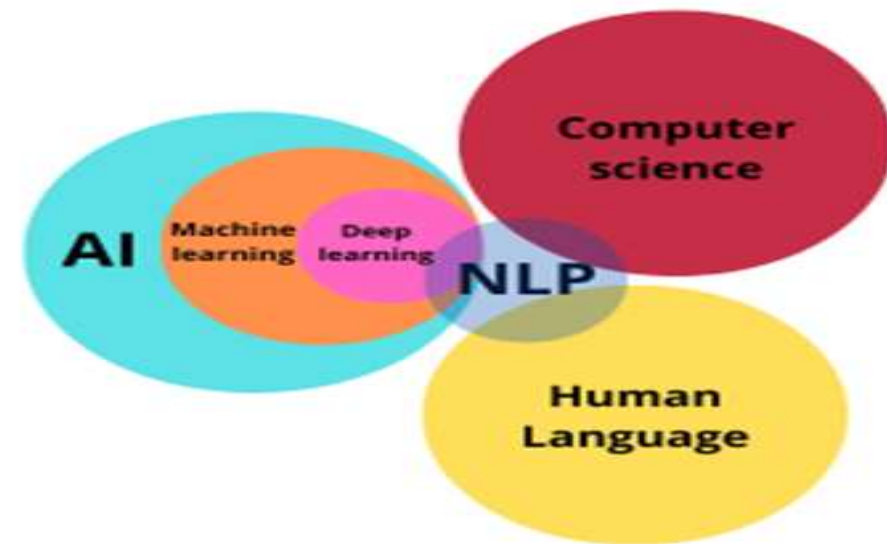


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Hours Per Week			Hours Per Semester			Credits	Assessment Marks		
L	T	P	L	T	P	C	CIE	SEE	Total
3	0	0	45	0	0	3	40	60	100

Introduction to Natural Language Processing

❖ **Natural language processing (NLP)** refers to the branch of computer science—and more specifically, the branch of artificial intelligence — concerned with giving computers the ability to understand text and spoken words in much the same way human beings can.



Introduction to Natural Language Processing

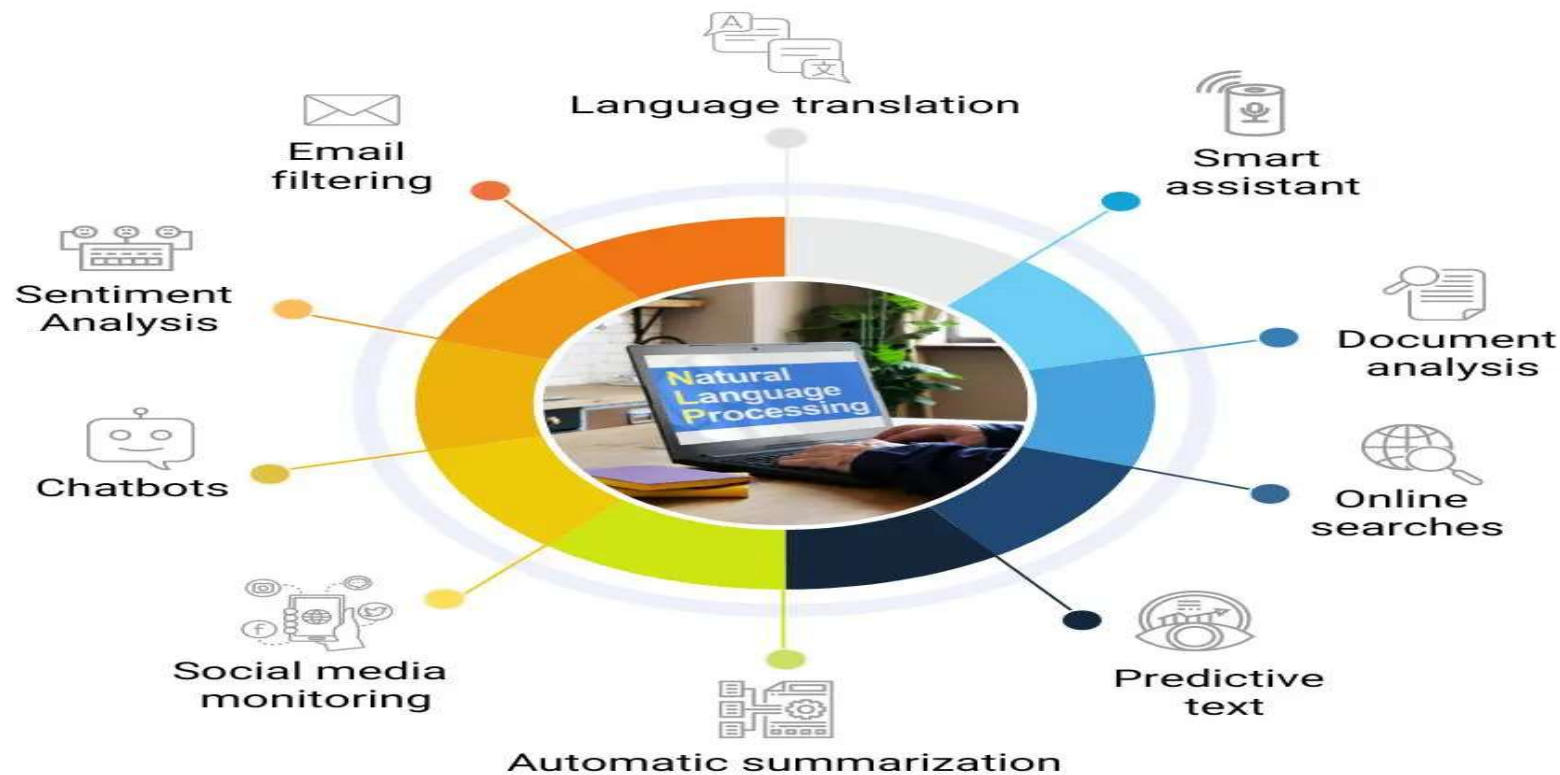


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- ❖ **NLP** combines computational linguistics—rule-based modeling of human language—with statistical, machine learning, and deep learning models.
- ❖ Together, these technologies enable computers to process human language in the form of text or voice data and to '**understand**' its full meaning, complete with the speaker or writer's intent and sentiment.
- ❖ **NLP** drives computer programs that translate text from one language to another, respond to spoken commands, and summarize large volumes of text rapidly—even in real time.

Applications of Natural Language Processing

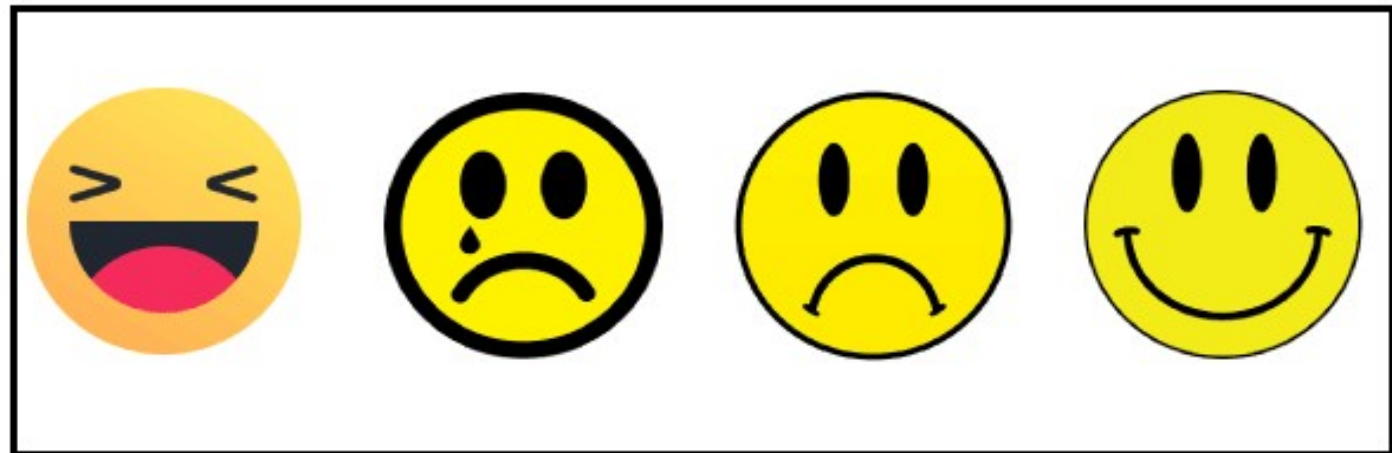
Applications of Natural Language Processing



Applications of Natural Language Processing

1. Sentiment Analysis

- ❖ Sentiment Analysis is also known as **opinion mining**.
- ❖ It is used on the web to analyse the attitude, behaviour, and emotional state of the sender.
- ❖ This application is assigning the values to the text (positive, negative, or natural), identify the mood of the context (happy, sad, angry, etc.)



Applications of Text Analytics

2 . Question Answering

Question Answering focuses on building systems that automatically answer the questions asked by humans in a natural language



3. Machine Translation

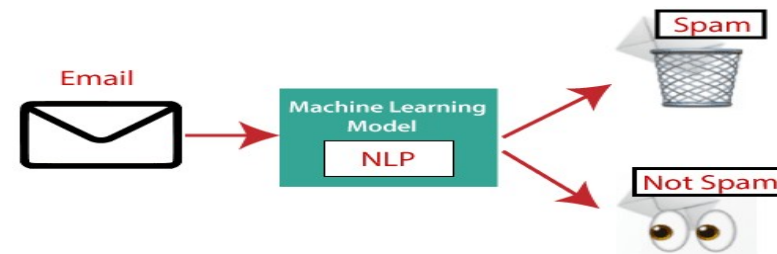
Machine translation is used to translate text or speech from one natural language to another natural language.

Example: Google Translator

Applications of Natural Language Processing

4. Spam Detection

Spam detection is used to detect unwanted e-mails getting to a user's inbox.



5. Spelling and Grammar correction

Microsoft Corporation provides word processor software like MS-word, PowerPoint for the spelling correction.

Example: Grammarly software

Applications of Natural Language Processing



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6. Speech Recognition

Speech recognition is used for converting spoken words into text. It is used in applications, such as mobile, home automation, video recovery, dictating to Microsoft Word, voice biometrics, voice user interface, and so on.

Example: google Assistant, Siri

7. Chatbot

Implementing the Chatbot is one of the important applications of NLP. It is used by many companies to provide the customer's chat services.

Applications of Natural Language Processing

8. Social media monitoring

It means tracking hashtags, keywords, and mentions relevant to your brand in order to stay informed about your audience and industry.



